

Potential Science Fair Projects

AI-driven AR safety-guidance research. Three dual-purpose ideas (ISEF ROBO/SOFT and IEEE VR/ISMAR poster). June 2026.

Each project is framed for an ISEF Engineering entry and a parallel XR research claim, in the standard **Pitch / Litmus Test** (**Novel, Feasible, Impactful, Testable**) structure. The target numbers in each Testable Goal are concrete, not placeholders.

Project Idea 1: “Safe-Fail,” Calibrated Abstention to Suppress False-Safe Errors in AI-Driven AR Hazard Guidance

Reframes AI-driven AR hazard guidance from an always-confident assistant into a system that knows when it is likely wrong and refuses to give a dangerous answer, positioning the work as a contribution in trustworthy, uncertainty-aware AI for safety-critical AR. (*Recommended lead project.*)

The Pitch: Vision-language models used for AR field guidance emit confident “Safe to Inspect” verdicts even when wrong, most dangerously on intact-looking but energized panels where a single false “safe” can be lethal. They optimize for average accuracy and have no usable sense of their own uncertainty. This project wraps the model in a calibrated-abstention “safe-fail” layer: it samples the model repeatedly (self-consistency / ensemble-disagreement), turns that disagreement into a calibrated confidence (temperature or Platt scaling against ground truth), and tunes the abstention threshold to the *asymmetric* cost of the domain, where a missed hazard is catastrophic but a false alarm only costs time. When confidence is low the overlay abstains (“Uncertain: verify manually”) instead of guiding a novice into danger. The contribution is not a better classifier; it is a system engineered to *fail safely*, plus the first evidence that surfacing calibrated uncertainty in an AR overlay changes whether novices catch the AI’s mistakes.

Litmus Test:

- **NOVEL:** Strong. Selective abstention for language and vision-language models exists (closest: CAP, arXiv 2502.06884), but it is generic and has never been applied to, or evaluated in, a safety-critical AR setting where a human acts on the verdict. The novelty is twofold: an *asymmetric-cost* abstention policy tuned to the lethal false-“safe” error rather than to overall accuracy, and the human-factors result that calibrated abstention in an overlay reduces novice over-reliance, a downstream link the abstention literature (which stops at the model’s output) has never measured.
- **FEASIBLE:** The lowest-risk of the three, for a structural reason: the core result comes from an *automated benchmark needing no human subjects and no IRB*. A solo student can build the sampling and calibration pipeline in Python around any VLM API, label a few-hundred-image set of intact versus hazardous components, and generate thousands of trials on their own clock. The human study is upside, not a dependency; if IRB or recruitment slips, the project still stands complete.
- **IMPACTFUL:** Two-fold. Practically, it fixes the most dangerous failure mode of AI field-guidance, confident wrong “safe” calls near lethal voltage, which is the gating obstacle to deploying such a tool to real apprentices. Scientifically, it contributes a reusable safe-fail pattern (asymmetric-cost abstention for human-facing AI) and the first evidence that uncertainty-aware overlays reduce automation bias, transferable to any high-stakes AI-assisted task (surgery, driving, inspection). “Zero hazards ever marked safe” is a memorable, judge-ready impact line.
- **TESTABLE: Testable Goal:** “To engineer a calibrated-abstention wrapper that reduces the false-‘safe’ rate (confident incorrect ‘Safe’ verdicts on hazardous components) by at least 50% relative to the uncalibrated baseline at matched coverage, achieving Expected Calibration Error below 0.10 and

abstention AUROC above 0.85; and to show in a controlled study (adult novices, N of 30 or more) that the calibrated/abstaining overlay increases the proportion of the AI's false-'safe' errors caught by users significantly ($p < 0.05$, logistic mixed-effects model) versus an always-confident overlay."

Project Idea 2: "Gaze-Lock," A Gaze-Contingent Verification Interlock for AR Safety Guidance

Reframes AR safety guidance from a passive overlay into a closed-loop system that uses the technician's own eyes as a safety sensor, refusing to confirm "safe" until the user has actually looked at the hazard, positioning it as a novel human-in-the-loop interaction technique for trustworthy AR.

The Pitch: An overlay that simply prints "Safe" trains novices to glance, trust, and proceed, the automation bias that makes AR guidance dangerous when the AI is wrong; worse, a highlight box can capture attention to the labeled region and suppress the independent search that would catch an *unlabeled* hazard. This project builds a gaze-contingent interlock: using a screen-based eye tracker (Tobii Pro Spark), it tracks where the novice looks and defers the "Safe / proceed" state until their gaze has dwelled on the hazard regions, turning verification from an assumption into a measured precondition, a cognitive forcing function in the perceptual loop rather than a dialog box. The contribution is coupling *verified gaze* to AI verdict-gating in AR, and showing it raises hazard fixation and error-catching in novices.

Litmus Test:

- **NOVEL:** A new coupling, with an honest caveat. AR attention-guidance techniques *direct* the eye (attention funnel, Biocca et al., CHI 2006; subtle cueing, Lu et al., IEEE TVCG), and cognitive forcing functions reduce over-reliance in non-AR tasks (Buçinca et al., CSCW 2021), but no prior system *gates an AI's safety verdict on verified fixation*. Caveat: the delta from that prior work is real but more incremental than Project 1, so the framing must foreground the closed-loop gating, not just "we measured gaze."
 - **FEASIBLE:** Medium, higher-risk than Project 1. Eye-tracker integration (calibration, region-of-interest dwell logic, real-time gating) is real engineering, and, more important, the statistical weight rests almost entirely on the human study because the gating logic is rule-based and generates little automated data, so the headline results are bottlenecked behind IRB approval and recruitment. On a solo 6–9 month, IRB-gated clock, that is the main threat to adequate power.
 - **IMPACTFUL:** Real but narrower than Project 1. It demonstrates a concrete interaction technique any heads-up safety system could adopt against automation bias, and "the system makes you prove you looked" is vivid to judges. But the impact is concentrated in the interaction-technique contribution rather than a reusable algorithm, and it depends on eye-tracking hardware, limiting near-term deployability on commodity phones.
 - **TESTABLE: Testable Goal:** "To engineer a gaze-contingent interlock that defers the 'Safe' verdict until verified fixation on hazard regions, and to show in a controlled study (adult novices, N of 30 or more, within-subjects) that, versus an immediate-verdict overlay, it significantly increases the proportion of the AI's false-'safe' errors caught ($p < 0.05$) and reduces time-to-first-fixation on the true hazard, while adding no more than a pre-specified ceiling of task time per step."
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Project Idea 3: "Two-Channel Safe-Fail," Confidence-Driven Gaze Verification

Unifies the two ideas above into one layered safe-fail architecture where the AI's own calibrated uncertainty decides when to demand that the human verify with their eyes, positioning the work as an adaptive, two-channel human-AI safety system that allocates scrutiny exactly where the AI is least sure. (*Highest ceiling, highest risk; best as a Year-2 continuation.*)

The Pitch: Calibrated abstention (Project 1) tells the system *when* it is likely wrong; gaze-gating (Project 2) forces the human to *verify*. Separately, each has a weakness: abstention can over-trigger and erode trust, and a universal gaze gate slows every step. This project fuses them, letting the calibrated confidence control how much verification to demand, clearing high-confidence-correct cases instantly but escalating low-confidence or “caution” cases to require gaze confirmation before the user proceeds. The result spends the user's scrutiny budget exactly where the AI is least reliable, maximizing caught errors at minimal added time. The contribution is the closed-loop integration of model uncertainty and human perceptual verification, two channels that cover each other's failure modes.

Litmus Test:

- **NOVEL:** The highest ceiling of the three. Neither the abstention nor the attention-guidance literature has built a system where calibrated model uncertainty *dynamically governs* a human-verification interlock. It is a genuinely new architecture, not combination for its own sake: the channels are complementary (the model catches errors the human's attention misses; the human catches errors the model is wrong-but-confident about). The strongest “Creativity and Potential Impact” story for ISEF.
- **FEASIBLE:** The lowest of the three, and the honest reason is scope against a fixed clock. It requires building *both* subsystems (calibration pipeline and eye-tracker interlock) and a multi-arm study (baseline, abstention-only, gaze-only, combined) to prove the integration earns its complexity, inflating the required sample size against an IRB-gated timeline. The realistic failure mode is an under-powered study that cannot separate the combined system from its parts. Best as a continuation once Projects 1 and 2 exist.
- **IMPACTFUL:** The highest ceiling: a blueprint for adaptive human-AI safety systems that allocate human oversight by machine confidence, applicable far beyond HVAC (clinical decision support, autonomous-vehicle handoffs, inspection). But the *incremental* real-world impact over Project 1 alone is modest until eye-tracking is commodity hardware, so its value is more conceptual than immediately deployable.
- **TESTABLE: Testable Goal:** “To engineer a two-channel safe-fail system in which calibrated abstention confidence triggers gaze-verified confirmation on uncertain cases, and to show in a controlled study (adult novices, N of 48 or more to power a four-arm comparison) that the integrated system catches significantly more false-‘safe’ errors than either calibrated abstention alone or gaze-gating alone ($p < 0.05$, planned contrasts), while adding no more than a pre-specified ceiling of task time per step.”

Ranking: Project 1 (lead), then Project 2, then Project 3 (stretch / continuation). Project 1 wins because its core results come from an automated benchmark independent of the IRB-gated human pipeline, giving a complete, statistically heavy project even in the worst case, while still yielding the IEEE VR / ISMAR poster. Run Project 2's gaze arm inside Project 1's study only if eye-tracker integration runs ahead of schedule; that path grows into Project 3 as a Year-2 continuation.

Compliance: all three need IRB / SRC pre-approval before any recruitment or data collection; recruit adult novices (18+) to avoid the minor assent plus parental-permission step; test only on de-energized mockups, photos, or video. Keep any conference submission anonymized (no identifying author, institution, or product branding) and present at the fair outside the conference review window.